

T0-Theory: Neutrinos The Photon Analogy, Geometric Oscillations, and Koide Extension

Document 5 of the T0 Series

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Koide extension: With the document's own masses 4.20/4.54/5.12 meV, $Q_\nu = 0.334 \approx 1/3$, not $2/3$; a quasi-degenerate spectrum always gives close to $1/3$. Moreover, the formula $\xi_0^{p_i+\delta} \nu_\nu$ gives a strong hierarchy, not these values.
- Mass differences from these masses: $\Delta m_{21}^2 = 0.030$, $\Delta m_{31}^2 = 0.086 \times 10^{-4} \text{ eV}^2$ instead of $(0.1-0.2) \times 10^{-4}$. With the measured Δm^2 , $\Sigma m_\nu \geq 59 \text{ meV}$; the spectrum with $\Sigma m_\nu = 13.86 \text{ meV}$ is excluded by oscillation data.
- Photon analogy: E^2 formula and cross section σ_ν corrected.
- Cosmology: non-relativistic at $z \approx 8$ instead of $z \sim 100$; $\Omega_\nu h^2 = 1.49 \times 10^{-4}$, $\Omega_\nu = 3.3 \times 10^{-4}$. Authoritative for neutrinos is Doc. 340 (addendum below): 4.54/9.81/49.9 meV, reproducing the measured Δm^2 .

Abstract

This document addresses the special position of neutrinos in the T0 Theory. In contrast to established particles (charged leptons, quarks, bosons), neutrinos require a fundamentally different treatment based on the photon analogy with double ξ_0 -suppression. The neutrino mass is derived from the formula $m_\nu = \frac{\xi_0^2}{2} \times m_e = 4.54 \text{ meV}$, and oscillations are explained by geometric phases based on $T_x \cdot m_x = 1$, where the quantum numbers (n, ℓ, j) determine the phase differences. An extension via the Koide relation introduces a weak hierarchy through exponent rotations; the resulting quasi-degenerate spectrum gives $Q_\nu \approx 1/3$ and is excluded by the measured mass differences (see correction notes). A plausible target value for the neutrino mass ($m_\nu = 15 \text{ meV}$) is derived from empirical data (cosmological limits). The T0 Theory is based on speculative geometric harmonies without empirical basis and is highly likely to be incomplete or incorrect. Scientific integrity requires a clear separation between mathematical correctness and physical validity.

.1 Preamble: Scientific Honesty

Warning

CRITICAL LIMITATION: The following formulas for neutrino masses are **speculative extrapolations** based on the untested hypothesis that neutrinos follow geometric harmonies and all flavor states have equal masses. This hypothesis has **no empirical basis** and is highly likely to be incomplete or incorrect. The mathematical formulas are nevertheless internally consistent and correctly formulated.

Scientific integrity means:

- Honesty about the speculative nature of the predictions
- Mathematical correctness despite physical uncertainty
- Clear separation between hypotheses and verified facts

.2 Neutrinos as “Almost Massless Photons”: The T0 Photon Analogy

Speculation

Fundamental T0 Insight: Neutrinos can be understood as “damped photons”. The remarkable similarity between photons and neutrinos suggests a deeper geometric kinship:

- **Speed:** Both propagate nearly at the speed of light
- **Penetration:** Both have extreme penetrability
- **Mass:** Photon exactly massless, neutrino quasi-massless
- **Interaction:** Photon electromagnetic, neutrino weak

Photon-Neutrino Correspondence

Physical Parallels:

$$\text{Photon: } E^2 = (pc)^2 + 0 \quad (\text{perfectly massless}) \quad (1)$$

$$\text{Neutrino: } E^2 = (pc)^2 + \left(\frac{\xi^2}{2} m_e c^2\right)^2 \quad (\text{quasi-massless}) \quad (2)$$

(Corrected on 30 Sept. 2026: previously $\sqrt{\xi^2/2} mc^2$; $\sqrt{\xi^2/2} = \xi/\sqrt{2} = 9.4 \times 10^{-5}$ contradicts this document's mass formula $m_\nu = \frac{\xi^2}{2} m_e$ which contains the factor 8.89×10^{-9} .)

Speed Comparison:

$$v_\gamma = c \quad (\text{exact}) \quad (3)$$

$$v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right) \approx 0.9999999911 \times c \quad (4)$$

The speed difference is only 8.89×10^{-9} – practically immeasurable!

The Double ξ_0 -Suppression**Key Result****Neutrino Mass through Double Geometric Damping:**

If neutrinos are "almost photons", then two suppression factors arise:

1. **First ξ_0 Factor:** "Almost massless" (like photon, but not perfect)
2. **Second ξ_0 Factor:** "Weak interaction" (geometric decoupling)

Resulting Formula:

$$m_\nu = \frac{\xi_0^2}{2} \times m_e = \frac{\left(\frac{4}{3} \times 10^{-4}\right)^2}{2} \times 0.511 \text{ MeV} \quad (5)$$

Numerical Evaluation:

$$m_\nu = 8.889 \times 10^{-9} \times 0.511 \text{ MeV} = 4.54 \text{ meV} \quad (6)$$

Physical Justification of the Photon Analogy**Why the Photon Analogy is Physically Sensible:****1. Speed Comparison:**

$$v_\gamma = c \quad (\text{exact}) \quad (7)$$

$$v_\nu = c \times \left(1 - \frac{\xi_0^2}{2}\right) \approx 0.9999999911 \times c \quad (8)$$

The speed difference is only 8.89×10^{-9} – practically immeasurable!

2. Interaction Strengths:

$$\sigma_\gamma \sim \alpha_{EM} \approx \frac{1}{137} \quad (9)$$

$$\sigma_\nu \sim \frac{\xi_0^2}{2} \times G_F \approx 1.04 \times 10^{-13} \text{ GeV}^{-2} \quad (10)$$

(Corrected on 30 Sept. 2026: previously $\approx 8.89 \times 10^{-9}$ and the sentence "The ratio $\sigma_\nu/\sigma_\gamma \sim \frac{\xi_0^2}{2}$ confirms the geometric suppression!". The number was only $\xi_0^2/2$; with $G_F = 1.166 \times 10^{-5} \text{ GeV}^{-2}$, σ_ν has the dimension GeV^{-2} , and comparing it with the dimensionless α_{EM} does not give a ratio $\xi_0^2/2$.)

3. Penetrability:

- Photons: Electromagnetic shielding possible
- Neutrinos: Practically unshieldable
- Both: Extreme ranges in matter

.3 Neutrino Oscillations

The Standard Model Problem

Warning

Neutrino Oscillations: Neutrinos can change their identity (flavor) during flight - a phenomenon known as neutrino oscillation. A neutrino produced as an electron neutrino (ν_e) can later be measured as a muon neutrino (ν_μ) or tau neutrino (ν_τ) and vice versa.

The oscillations depend on the mass squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$ and the mixing angles. Current experimental data (2025) provide:

$$\Delta m_{21}^2 \approx 7.53 \times 10^{-5} \text{ eV}^2 \quad [\text{Solar}] \quad (11)$$

$$\Delta m_{32}^2 \approx 2.44 \times 10^{-3} \text{ eV}^2 \quad [\text{Atmospheric}] \quad (12)$$

$$\Sigma m_\nu \gtrsim 0.06 \text{ eV} \quad [\text{from oscillation data; heaviest neutrino} \gtrsim 0.05 \text{ eV}] \quad (13)$$

Problem for T0: The T0 Theory postulates equal masses for the flavor states (ν_e, ν_μ, ν_τ), which implies $\Delta m_{ij}^2 = 0$ and is incompatible with standard oscillations.

Geometric Phases as Oscillation Mechanism

Speculation

T0 Hypothesis: Geometric Phases for Oscillations

To reconcile the hypothesis of equal masses ($m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu$) with neutrino oscillations, it is speculated that oscillations in the T0 Theory are caused by geometric phases rather than mass differences. This is based on the T0 relation:

$$T_x \cdot m_x = 1,$$

where $m_x = m_\nu = 4.54 \text{ meV}$ is the neutrino mass and T_x is a characteristic time or frequency:

$$T_x = \frac{1}{m_\nu} = \frac{1}{4.54 \times 10^{-3} \text{ eV}} \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s}.$$

The geometric phase is determined by the T0 quantum numbers (n, ℓ, j) :

$$\phi_{\text{geo},i} \propto f(n, \ell, j) \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f(n, \ell, j) = \frac{n^6}{\ell^3}$ (or 1 for $\ell = 0$) are the geometric factors:

$$f_{\nu_e} = 1, \quad (14)$$

$$f_{\nu_\mu} = 64, \quad (15)$$

$$f_{\nu_\tau} = 91.125. \quad (16)$$

WARNING: This approach is purely hypothetical and without empirical confirmation. It contradicts the established theory that oscillations are caused by $\Delta m_{ij}^2 \neq 0$.

Quantum Number Assignment for Neutrinos

Neutrino Flavor	n	ℓ	j	$f(n, \ell, j)$
ν_e	1	0	1/2	1
ν_μ	2	1	1/2	64
ν_τ	3	2	1/2	91.125

Table 1: Speculative T0 Quantum Numbers for Neutrino Flavors

.4 Integration der Koide-Relation: Eine schwache Hierarchie

T0-Koide Extension for Neutrinos:

To address the oscillation conflict ($\Delta m_{ij}^2 \neq 0$), the T0 Theory integrates the Koide relation as a natural generalization (Li and Ma 2005; Brannen 2006). This introduces a weak hierarchy via exponent rotations around ξ_0 , preserving the photon analogy while enabling small mass differences.

Eigenvector Representation: The charged lepton masses follow Koide via:

$$\begin{pmatrix} \sqrt{m_e} \\ \sqrt{m_\mu} \\ \sqrt{m_\tau} \end{pmatrix} = \mathbf{U} \cdot \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}, \quad (17)$$

where $\mathbf{U}$ is the unitary flavor-mixing matrix (CKM/PMNS analog).

T0 Adaptation for Neutrinos: Neutrino masses emerge as perturbed versions of the base $m_\nu = 4.54 \text{ meV}$:

$$m_{\nu_i} \approx \xi_0^{p_i + \delta} \cdot \nu_\nu, \quad \delta \approx \xi_0^{1/3} \approx 0.051 \quad (18)$$

with exponents $p_i = (3/2, 1, 2/3)$ from charged leptons (rotated by δ for weak hierarchy). This yields a quasi-degenerate spectrum:

$$m_{\nu_1} \approx 4.20 \text{ meV (normal hierarchy),} \quad (19)$$

$$m_{\nu_2} \approx 4.54 \text{ meV,} \quad (20)$$

$$m_{\nu_3} \approx 5.12 \text{ meV,} \quad (21)$$

$$\Sigma m_\nu \approx 13.86 \text{ meV.} \quad (22)$$

(Note of 30 Sept. 2026: The formula $\xi_0^{p_i+\delta} \nu_i$ with $p_i = (3/2, 1, 2/3)$ gives $m_{\nu_1}/m_{\nu_3} = \xi_0^{3/2-2/3} = 5.9 \times 10^{-4}$, i.e. a strong hierarchy as for the charged leptons, not the quasi-degenerate values stated.)

Neutrino Koide Relation:

$$Q_\nu = \frac{m_{\nu_1} + m_{\nu_2} + m_{\nu_3}}{(\sqrt{m_{\nu_1}} + \sqrt{m_{\nu_2}} + \sqrt{m_{\nu_3}})^2} = \frac{13.86}{41.51} \approx 0.334, \quad (23)$$

i.e. close to $1/3$, the value for exactly equal masses. (Corrected on 30 Sept. 2026: previously " $\approx 0.6667 = \frac{2}{3}$, with $\Delta Q_\nu < 1\%$ accuracy, directly linking to PMNS mixing". For a quasi-degenerate spectrum Q_ν tends to $1/3$; $2/3$ cannot be reached this way.)

Hybrid Oscillation Mechanism: Geometric phases (from $f(n, \ell, j)$) dominate, augmented by small $\Delta m_{21}^2 \approx 0.030 \times 10^{-4}$ and $\Delta m_{31}^2 \approx 0.086 \times 10^{-4} \text{ eV}^2$ from δ . (Corrected on 30 Sept. 2026: previously $\Delta m_{ij}^2 \approx (0.1 - 0.2) \times 10^{-4} \text{ eV}^2$ and "This reconciles T0 with data without full hierarchy". The values follow from 4.20/4.54/5.12 meV. With the measured $\Delta m_{21}^2 = 7.53 \times 10^{-5}$ and $\Delta m_{32}^2 = 2.44 \times 10^{-3} \text{ eV}^2$, $\Sigma m_\nu \geq 59 \text{ meV}$; the spectrum with 13.86 meV is excluded by the oscillation data, also by falsification criterion 3 below. Authoritative: Doc. 340.)

WARNING: Highly speculative; testable via future Σm_ν measurements (e.g., Euclid 2026+).

.5 Experimental Assessment

Cosmological Limits

Experimental

Cosmological Neutrino Mass Limits (as of 2025):

1. DESI + CMB (2024):

$$\Sigma m_\nu < 0.07 \text{ eV} \quad (95\% \text{ Confidence}) \quad (24)$$

2. T0 Prediction (with Koide Extension):

$$\Sigma m_\nu = 13.86 \text{ meV} \quad (25)$$

3. Comparison:

$$\frac{13.86 \text{ meV}}{70 \text{ meV}} = 0.198 \approx 19.8\% \quad (26)$$

The T0 prediction is well below all cosmological limits! (Note of 30 Sept. 2026: It is thereby also below the lower bound $\Sigma m_\nu \geq 59 \text{ meV}$ from oscillation data and is excluded, see section Koide relation.)

Direct Mass Determination

Experimental

Experimental Neutrino Mass Determination:

1. KATRIN Experiment (2022):

$$m(\nu_e) < 0.8 \text{ eV} \quad (90\% \text{ Confidence}) \quad (27)$$

2. T0 Prediction (with Koide):

$$m(\nu_e) \approx 4.54 \text{ meV (effective)} \quad (28)$$

3. Comparison:

$$\frac{4.54 \text{ meV}}{800 \text{ meV}} = 0.0057 \approx 0.57\% \quad (29)$$

The T0 prediction is orders of magnitude below the direct mass limits.

Target Value Estimation

Key Result

Plausible Target Value for Neutrino Masses:

From cosmological data and theoretical considerations, a plausible target value emerges:

$$m_\nu^{\text{Target}} \approx 15 \text{ meV (per flavor, quasi-degenerate)} \quad (30)$$

Comparison with T0 Prediction (incl. Koide):

$$\frac{4.54 \text{ meV}}{15 \text{ meV}} = 0.303 \approx 30.3\% \quad (31)$$

The T0 prediction is about a factor of 3 below the plausible target value, which is acceptable for a speculative theory. Koide extension narrows this to 7% via hierarchy.

.6 Cosmological Implications

Structure Formation and Big Bang Nucleosynthesis

Key Result

Cosmological Consequences of T0 Neutrino Masses:

1. Big Bang Nucleosynthesis:

- Relativistic neutrinos at $T \sim 1$ MeV: Standard BBN unchanged
- Contribution to radiation density: $N_{\text{eff}} = 3.046$ (Standard)

2. Structure Formation:

- Neutrinos with 4.5 meV become non-relativistic at $z \approx 8$ (*corrected on 30 Sept. 2026: previously $z \sim 100$; $1 + z \approx m_\nu / (3.15 T_{\nu,0}) = 4.5 / 0.53$ with $T_{\nu,0} = 1.68 \times 10^{-4}$ eV*)
- Suppression of small-scale structure formation negligible

3. Cosmic Neutrino Background (C ν B):

- Number density: $n_\nu = 336 \text{ cm}^{-3}$ (unchanged)
- Energy density: $\rho_\nu \propto \Sigma m_\nu = 13.86 \text{ meV}$ (with Koide)
- Fraction of critical density: $\Omega_\nu h^2 = 13.86 \text{ meV} / 93.14 \text{ eV} \approx 1.49 \times 10^{-4}$

4. Comparison with Dark Matter:

- Neutrino contribution: $\Omega_\nu \approx 3.3 \times 10^{-4}$ (with $h = 0.674$)
 - Dark matter: $\Omega_{DM} \approx 0.26$
 - Ratio: $\Omega_\nu / \Omega_{DM} \approx 1.3 \times 10^{-3}$ (negligible)
- (Corrected on 30 Sept. 2026: previously $\Omega_\nu h^2 \approx 1.55 \times 10^{-4}$, $\Omega_\nu \approx 2.1 \times 10^{-4}$ and $\Omega_\nu / \Omega_{DM} \approx 8.1 \times 10^{-4}$; the first two figures were only compatible with $h = 0.86$.)

.7 Experimental Tests and Falsification

Testable Predictions

Experimental

Specific Experimental Tests of the T0 Neutrino Theory:

1. Direct Mass Determination:

- KATRIN: Sensitivity to ~ 0.2 eV (insufficient)
- Future Experiments: ~ 0.01 eV required
- T0 Prediction: $m_{\nu_i} \approx 4 - 5$ meV (factor 2 below limit)

2. Cosmological Precision Measurements:

- Euclid Satellite: Sensitivity ~ 0.02 eV
- T0 Prediction: $\Sigma m_\nu = 13.86$ meV (testable!)

3. Koide-Specific Tests:

- Measure Q_ν via oscillation data: Expect $\approx 2/3$ ($\Delta < 1\%$) (*note of 30 Sept. 2026: this document's quasi-degenerate spectrum gives $Q_\nu \approx 1/3$*)
- PMNS correlations: Hierarchy from δ -rotation

4. Speed Measurements:

- Supernova Neutrinos: $\Delta v/c \sim 10^{-8}$ measurable
 - T0 Prediction: $\Delta v/c = 8.89 \times 10^{-9}$ (marginal)
- 5. Oscillation Physics:**
- Test for small Δm_{ij}^2 + phase effects (clearly falsifiable)

Falsification Criteria

The T0 Neutrino Theory would be falsified by:

1. Direct measurement of $m_\nu > 0.1$ eV (or strong hierarchy $|m_3 - m_1| > 10$ meV)
2. Cosmological evidence for $\Sigma m_\nu > 0.1$ eV
3. Clear proof of $\Delta m_{ij}^2 \gg 10^{-4}$ eV² without phases
4. Measurement of speed differences $\Delta v/c > 10^{-8}$
5. Deviation from $Q_\nu \approx 2/3$ in oscillation analyses (*note of 30 Sept. 2026: see section Koide relation, $Q_\nu \approx 1/3$*)

Addendum (3 September 2026, Doc. 340 and Doc. 343 Theorem D''')

The base formula $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54$ meV agrees exactly with m_{ν_1} in Doc. 340, which derives the neutrino masses algebraically from the GF(27)* Galois orbits **[K]**. Doc. 007 thereby provides the physical motivation for the double ξ suppression (neutrino = almost photon + geometric decoupling) that Doc. 340 treats only algebraically. Neither ξ nor m_e are free parameters: ξ follows from Galois group orders **[K]** (Doc. 338), m_e from torus quantum numbers **[K]** (Doc. 006); $\nu = 246$ GeV is an SI scale fixation, not a free parameter.

What Doc. 340 additionally achieves **[B]**:

- Three different masses from orbits O_1, O_3, O_4 : $m_{\nu_1} = 4.54$, $m_{\nu_2} = 9.81$, $m_{\nu_3} = 49.9$ meV — Δm_{21}^2 and Δm_{31}^2 reproduced to 0.4 and 1.5 %.
- Majorana/Dirac partition: ν_1, ν_2 Majorana (orbits O_1, O_3 self-inverse); ν_3 **Dirac neutrino** (orbit O_4 , Doc. 343 Theorem D''').
- No Majorana phase for ν_3 ; $m_{ee} \leq 14.4$ meV; inverted hierarchy more strongly excluded.

The Koide extension and the $n/\ell/j$ quantum numbers have no connection to the Galois orbits and remain **[S]**. Authoritative current state: Doc. 340 and Doc. 343.

.8 Limits and Open Questions

Fundamental Theoretical Problems

Warning

Unsolved Problems of the T0 Neutrino Theory:

1. **Oscillation Mechanism:** Geometric phases + δ are ad hoc
2. **Quantum Field Theory:** No complete QFT formulation
3. **Experimental Distinguishability:** Difficult to separate from Standard Model
4. **Theoretical Consistency:** Partial contradiction to oscillation theory

5. **Predictive Power:** Enhanced by Koide, but still limited

.9 Methodological Reflection

Scientific Integrity vs. Theoretical Speculation

Key Result

Central Methodological Insights:

The neutrino chapter of the T0 Theory illustrates the tension between:

- **Theoretical Completeness:** Desire for unified description (now incl. Koide)
- **Empirical Anchoring:** Necessity of experimental confirmation
- **Scientific Honesty:** Disclosure of speculative nature
- **Mathematical Consistency:** Internal self-consistency of formulas

Key Insight: Even speculative theories can be valuable if their limits are honestly communicated.

Significance for the T0 Series

The neutrino treatment shows both the strengths and limits of the T0 Theory:

- **Strengths:** Unified framework, elegant analogies, testable predictions (enhanced by Koide)
- **Limits:** Speculative basis, lack of experimental confirmation
- **Scientific Value:** Demonstration of alternative thinking approaches
- **Methodological Importance:** Importance of honest uncertainty communication

*This document is part of the new T0 Series
and shows the speculative limits of the T0 Theory*

T0-Theory: Time-Mass Duality Framework

Johann Pascher

GitHub: <https://github.com/jpascher/T0-Time-Mass-Duality>

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